

◆ M001 · Pyramid Variations — English Worksheet (Beginner)

Topic: Pyramid Variations · **Course:** Pyramid Problems · **Level:** Beginner · **Time:** about 30 minutes

A **hollow** layer has blocks only on the edge. A **stepped** pyramid shrinks faster, so the steps look bigger.

1 · Predict 🧠

Read the steps. Circle your answer.

```
set size to 5
while size > 0:
    [build one HOLLOW size × size layer]
    move up by 1
    set size to size - 2
```

Is this pyramid filled inside or empty inside? Circle one: filled · empty

Could a person stand inside it? Circle one: yes · no

```
set size to 9
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 4
```

This shrinks by 4, not 2. Do the steps look bigger or smaller? Circle one: bigger · smaller

2 · Find the Difference 🐛

Clean steps first, then a buggy version. Circle the bug.

Pair A — The layer should be **hollow** — blocks only on the edge.

```
# clean
[build one HOLLOW layer]
```

```
# buggy
[build one FULL layer]
```

What is wrong? Circle one: it is solid (filled) inside · it is empty inside · it has no walls

Pair B — The stepped pyramid should shrink by **4** each layer.

```
# clean
while size > 0:
    [build layer]
    move up by 1
    set size to size - 4
```

```
# buggy
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

What is wrong? Circle one: it shrinks by 2, not 4 · it shrinks by 4, not 2 · it does not shrink

3 · Fill the Gap

You want a **stepped** pyramid that shrinks fast. Fill the missing number.

```
set size to 9
while size > 0:
    [build layer]
    move up by 1
    set size to size - ____
```

Missing number? Circle one: 4 · 0 · 100

4 · Show Your Work 📷 🎥

Switch to your homework world. Pick **one** variation and build it:

- **Hollow** — layers with blocks only on the edge, or
- **Stepped** — shrink by 4 instead of 2.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.